Linearity preparation: row overscan
Mathematics and implementation, version 1

Wenu-imaging

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Scope and configuration

This guide describes the new in-memory preparation services, not a completed FITS-to-ASDF recipe or an accepted physical calibration. Actual Mac qualification and real corrected-ROI products remain separate gates. The reviewed design is in the developer design directory:

`linearity-preparation-overscan-asdf-v1.md`.

The detector TOML describes the full detector bands and dimensions. The processing TOML selects the sample columns, trim rectangle, rejection counts, ROIs and preparation controls. The shared Python resolver validates their relationship and freezes both original texts and SHA256 identities. There is no PAF fallback. For CCD65, the full overscan is columns 2097–2144, but the sample is 2100–2142. These distinctions survive in the resolved record.

Exact row correction and arithmetic

For raw frame f, raw row y, and the n enabled scan values sorted in ascending order, the existing trimmed-mean routine returns

$$O_f(y) = \mathrm{pix}\left[\frac{\sum_{k=l+1}^{n-h} v_{f,y,(k)}}{n-l-h}\right]. \tag{1}$$

The sum and division use double precision; pix converts to the native `pixelvalue` type. Low/high rejection is $l = h = 10$ in the selected OmegaCAM profile: 43 reads retain 23, and 44 retain 24. No sigma clipping, smoothing, row fitting, endpoint extrapolation or global-level replacement is introduced. First and last raw rows use their own scan samples. The active-pixel BPM is not applied to electronic scan reads.

The native stages are

$$U_f(x,y) = \mathrm{pix}[D_f(x,y) - O_f(y)], \tag{2}$$

$$S_f(p) = \mathrm{pix}[U_f(p) - B(p)] \quad \text{(flats only)}. \tag{3}$$

The trim uses inclusive FITS coordinates after scan subtraction. An active row j maps to raw row $j + y_{\min} - 1$. Preserve both native roundings: do not merge the subtractions. A supplied master is already scan-corrected and active; it is subtracted once and receives no second scan correction. Bias-role frame preparation has no master subtraction. The existing bias service performs frame admission and stacking; this increment does not change its median or rejection policy. Master lineage compatibility must be verified by the future file/recipe admission layer, not inferred from its pixel values.

Corrected ROI statistics and populations

The paired service first reuses the unchanged raw-pair measurement service. For each configured ROI it uses the common population $\mathcal{P}$ admitted by the declared BPM and finite/raw-saturation checks in both flats and both raw biases. The raw saturation boundary remains 65534 ADU. Pair members retain their original identity, HDU, input index and exact duration.

For corrected member values S_{1p}, S_{2p}, let $N = |\mathcal{P}|$. The services retain ordinary means and sample scatter of each member and their difference:

$$\overline{S}_i = N^{-1} \sum_{p \in \mathcal{P}} S_{ip}, \tag{4}$$

$$s_i^2 = (N-1)^{-1} \sum_{p \in \mathcal{P}} (S_{ip} - \overline{S}_i)^2, \tag{5}$$

$$d_p = \mathrm{pix}[S_{1p} - S_{2p}], \qquad s_d^2 = (N-1)^{-1} \sum_{p \in \mathcal{P}} (d_p - \overline{d})^2. \tag{6}$$

Difference pixel arithmetic follows `image_pair_stats_checked`. The native result stores standard deviations and availability; a later serialization stage may store their squared values and $s_d^2/2$ with explicit definitions. For $N < 2$, sample scatter is unavailable, not zero. A corrected nonfinite value cannot silently remove an otherwise admitted measured pixel: all three corrected population counts must equal the raw count.

Spatial scatter and difference scatter are diagnostic quantities, not assigned independent weights. The shared master cancels algebraically in differences, subject to native rounding. Raw bias-only RON and raw gain diagnostics retain their original definitions. No uncertainty model or physical response is estimated. ROI means remain distinct from the direct physical pixel samples required by the later nonlinear model.

Connection to implementation and verification

`pycavi_linearity_config.resolve` performs TOML resolution only.

`mef_linearity_prepare_frame` validates geometry and finite enabled scan reads, copies the borrowed raw image and uses the existing `mef_overscan_correction_traced` arithmetic. Its optional trace copies each estimate before QC median sorting, preserving physical row order. The legacy interfaces pass a null trace and keep their numerical behavior. The result owns the active image, trace and settings copy.

`mef_linearity_prepare_pairs` corrects the two biases for trace evidence and two flats at a time, reusing raw-pair admission and the existing masked single-image and difference statistics. Results own copied identities, raw diagnostics, corrected ROI statistics and traces. Borrowed inputs stay unchanged; occupied outputs, invalid settings, resource limits and failures do not expose partial completed owners. Custom allocation checks cover the new owners and snapshot owners; they do not establish general Eclipse allocation safety.

Independent ramp fixtures cover both scan widths, raw/active edges, exact singleton-path agreement, common-population analytical moments, and persistent owner-allocation failures. Existing bias-service, sample and raw-pair controls check affected shared paths. Six TOML and five footprint controls check the configuration boundary. Agent Linux success is not actual Mac acceptance or scientific validation. Complete recipe/ASDF orchestration follows that gate.